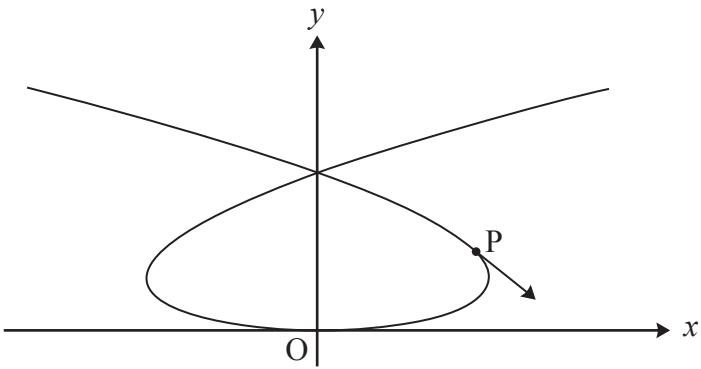
8 A particle moves in the x-y plane so that its position at time t s is given by $x = t^3 - 8t$, $y = t^2$ for $-3.5 < t < 3.5$. The units of distance are metres. The graph shows the path of the particle and the direction of travel at the point P (8, 4).



(a) Find $\frac{dy}{dx}$ in terms of t. **[3]**

(b) Hence show that the value of $\frac{dy}{dx}$ at P is -1. **[2]**

(c) Find the time at which the particle is travelling in the direction opposite to that at P. **[2]**

(d) Find the cartesian equation of the path, giving x^2 as a function of y. **[3]**